NEET Physics

Detailed NEET-UG Syllabus

Exam Coverage	NEET-UG aligned topic map for complete subject planning
Format	Unit-wise syllabus with detailed bullet-point subtopic breakdown
Use Case	Course brochure, batch syllabus design, chapter planning and academic structuring

Structured Syllabus

Unit 1: Physics and Measurement

- Units and dimensions, SI system, dimensional analysis and conversion of units.
- Errors in measurement, significant figures, precision and accuracy, least count and graphical representation of data.

Unit 2: Kinematics

- Motion in one and two dimensions, position, displacement, distance, speed, velocity and acceleration.
- Graphical analysis of motion, equations of uniformly accelerated motion, projectile motion and uniform circular motion.

Unit 3: Laws of Motion

- Newton’s laws of motion, inertia, momentum, impulse and applications of force.
- Static and kinetic friction, free body diagrams, connected systems and circular motion dynamics.

Unit 4: Work, Energy and Power

- Work done by constant and variable forces, kinetic and potential energy, work-energy theorem.
- Conservation of mechanical energy, power, collisions and centre of mass related applications.

Unit 5: Rotational Motion

- Centre of mass, moment of inertia, radius of gyration, torque and angular momentum.
- Rolling motion, equilibrium of rigid bodies and rotational dynamics in simple systems.

Unit 6: Gravitation

- Universal law of gravitation, acceleration due to gravity and variation of g.
- Gravitational potential, escape velocity, orbital motion and satellites.

Unit 7: Properties of Bulk Matter

- Elasticity, stress-strain relation, Young’s modulus and applications.
- Pressure in fluids, viscosity, streamline flow, Bernoulli’s theorem, surface tension and capillarity.

Unit 8: Thermodynamics

- Thermal equilibrium, heat, temperature, calorimetry and transfer of heat.
- First law of thermodynamics, isothermal and adiabatic processes, heat engines and refrigerators.

Unit 9: Kinetic Theory of Gases

- Equation of state of a perfect gas, kinetic interpretation of temperature, RMS speed and degrees of freedom.
- Specific heat capacities and relation between microscopic and macroscopic properties.

Unit 10: Oscillations and Waves

- Simple harmonic motion, spring-mass systems, pendulum, energy in SHM, resonance and damping.
- Wave motion, sound waves, Doppler effect, standing waves and beats.

Unit 11: Electrostatics

- Electric charges, Coulomb’s law, electric field, electric flux and Gauss’s law.
- Electric potential, equipotential surfaces, potential energy and capacitors.

Unit 12: Current Electricity

- Electric current, drift velocity, Ohm’s law, resistivity and electrical circuits.
- Series and parallel combinations, cells, Kirchhoff’s laws, Wheatstone bridge and potentiometer.

Unit 13: Magnetic Effects of Current and Magnetism

- Magnetic field due to current, Biot–Savart law, Ampere’s law and force on charges and conductors.
- Magnetic properties of matter, Earth’s magnetism and moving charge in magnetic field.

Unit 14: Electromagnetic Induction and Alternating Current

- Faraday’s law, Lenz’s law, induced EMF, self and mutual induction.
- Alternating current, reactance, impedance, resonance and transformers.

Unit 15: Electromagnetic Waves

- Displacement current and electromagnetic waves, their characteristics and spectrum.

Unit 16: Optics

- Reflection and refraction, mirrors and lenses, lens maker’s formula and optical instruments.
- Wave optics: interference, diffraction, Young’s double slit experiment and polarization.

Unit 17: Dual Nature of Matter and Radiation

- Photoelectric effect, Einstein’s equation, de Broglie wavelength and matter waves.

Unit 18: Atoms and Nuclei

- Atomic models, Bohr model, line spectra, nuclear composition, radioactivity and nuclear reactions.

Unit 19: Electronic Devices

- Semiconductors, intrinsic and extrinsic types, p-n junction, diode and transistor basics.
- Logic gates, digital electronics and common electronic applications.